**VISVESVARAYA TECHNOLOGICAL UNIVERSITY BELAGAVI – 590018**

Project Based Assignment
Report on

# GAS LEAKAGE SYSTEM

[BEC405A-Microcontrollers 8051]

Submitted in partial fulfillment for the award of degree

**BACHELOR OF ENGINEERING**

in

**ELECTRONICS AND COMMUNICATION ENGINEERING**

By

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**Academic Year 2024-25**

# Vision

To be an Institute of Excellence in Engineering education, Innovation, Research and work towards evolving great leaders for the country's future and meeting global needs.

# Mission

The Institution aims at providing a vibrant, intellectually and emotionally rich teaching learning environment with the State of the Art Infrastructure and recognizing and nurturing the potential of each individual to evolve into one's own self and contribute to the welfare of all.

## Vision

To be a Department of excellence at a global level in Electronics and Communication Engineering education, incorporating Research & Innovation and Leadershi training components.

## Mission

M1. Strive to provide state of Art infrastructure in classrooms and laboratories.

M2. Enable all-round development with individual attention and innovative teaching learning methodology.

M3. Impart leadership qualities into the students by exposing them to industry anc research in global Electronics and Communication Engineering domain.

## Programme Ec ucation Objectives

PEO1. (Domain Knowledge) Graduates of Electronics & Communication Engineering will be able to utilize mathematics, science, engineering fundamentals, theoretical as well as laboratory based experiences to identify, formulate & solve engineering problems and succeed in advanced engineering or other fields.

PEO2. (Professional Employment) Graduates of Electronics & Communication Engineering will succeed in entry-level engineering positions in VLSI, Communication and Fabrication industries in regional, national, or global industries.

PEO3. (Engineering Citizenship) Graduates of Electronics & Communication Engineering will be prepared to communicate and work effectively on individual & team based engineering projects while practicing the ethics of their profession consistent with a sense of social responsibility.

PEO4. (Lifelong Learning) Graduates of Electronics & Communication Engineering will be equipped to recognize the importance of, and have the skills for, continuous learning to become experts in their domain and enhance their professional attributes.

## Programme Specific C utcomes

PSO1. (Knowledge / Skills) Explore emerging technologies in the field of Electronics & Communication Engineering using the knowledge and skills gained.

PSO2. (Application/Analysis/Problem solving) Apply techniques in different domains to create innovative products and services in the Communication, VLSI, DSP, and Networking.

PSO3. (Value/ Attribute) Work on various platforms as an individual/ team member to develor useful and safe Circuits, PCB, Power Management Systems anc Automation for the society and nation.

# ABSTRACT

This project presents a gas leakage detection and monitoring system designed to enhance safety in environments where gases like LPG, methane, and carbon monoxide are used. The system employs high-sensitivity gas sensors capable of detecting various gases and continuously monitoring their concentration levels in real-time. Upon detecting a gas leak, the system triggers local audio and visual alarms and simultaneously sends notifications to designated personnel through SMS or a mobile application, ensuring immediate awareness and response. Additionally, the system features data logging and visualization capabilities, maintaining a historical record of gas readings and displaying them through intuitive dashboards for analysis and trend monitoring. Remote access functionality allows users to oversee gas levels and manage the system from a distance, offering convenience and control. The benefits of this system include enhanced safety through early leak detection, improved emergency response times, reduced property damage and financial losses, and increased user awareness and control over gas usage. Its adaptability makes it suitable for various applications such as homes, industries, and public facilities. Overall, this project aims to develop a reliable and cost-effective solution that significantly contributes to safety and well-being in gas-utilizing environments.

# 1. INTRODUCTION

## 1.1 Background

In various industries and medical applications, temperature control plays a crucial role in ensuring safety, comfort, and proper functioning of equipment. One such application is temperature optimization. Traditional thermostat or analog-based temperature control systems often suffer from limited accuracy and sensitivity. As the need for more reliable and precise systems grows, digital temperature control solutions using advanced sensors and microcontroller technology have become a key area of focus.

## 1.2 Problem Statement

Gas leaks pose a significant threat to safety and property, with potential for explosions, fires, and toxic exposure. Early detection and response are crucial to minimize damage and save lives.

## 1.3 Objectives

design and develop a Precise Digital Temperature Controller for applications requiring strict temperature regulation

The system aims to provide highly accurate temperature measurements and control, offering a more reliable solution compared to conventional thermostat systems.

The system will automatically regulate the temperature by switching off the heater when it exceeds preset limits, ensuring the temperature stays within the desired range for optimal care and comfort.

## 1.4 Scope of the Project

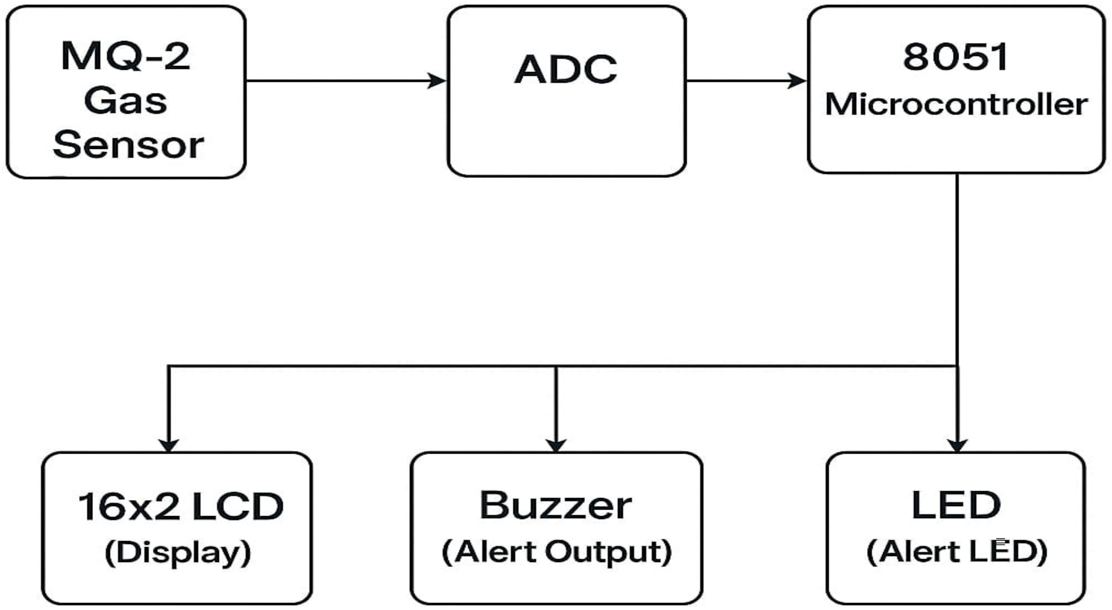
The project focuses on using the 89C51 microcontroller with a gas sensor and ADC0808 to build a gas leak detection system suitable for small-scale deployment in residential kitchens, laboratories, or storage areas. It emphasizes real-time monitoring of hazardous gases, immediate alert generation through alarms or indicators, and reliable sensor integration to ensure user safety and system efficiency.

# 2. LITERATURE SURVEY

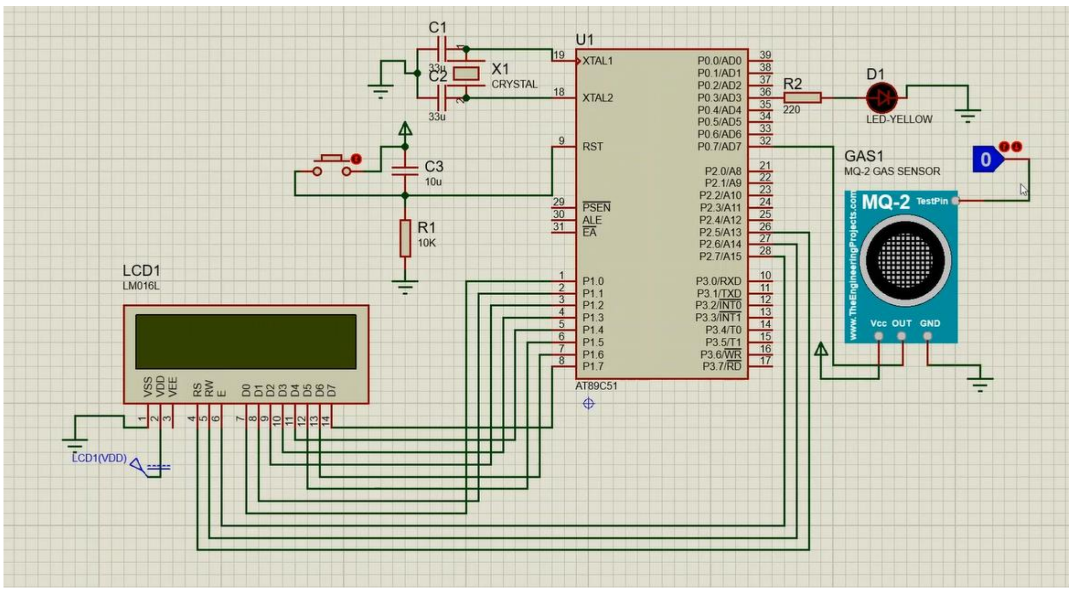
Several gas leakage detection systems have been developed using modern microcontrollers such as Arduino, Raspberry Pi, and ESP32, often integrated with advanced communication modules for IoT applications. These systems typically utilize MQ-series gas sensors and provide cloud-based or mobile app alerts. However, limited literature is available on gas leak detection systems implemented using the 89C51 microcontroller along with ADC0808 and basic sensors, particularly in combination with simple buzzer alert mechanisms. Most existing systems either rely on expensive components or introduce complexity that is unsuitable for small-scale or educational use. Our project addresses this gap by presenting a low-cost, easy-to-build gas leak detection system using the 89C51 microcontroller, ADC0808, and a suitable gas sensor, providing efficient real-time alerts through a buzzer. This approach makes the system ideal for deployment in homes, laboratories, and other confined spaces where simplicity, affordability, and reliability are essential.

# 3.SYSTEM DESIGN AND METHODOLOGY

## 3.1 Block diagram




## 3.2 Circuit Diagram



## 3.3 Description of Components

- **89C51 Microcontroller:** A classic 8051 family microcontroller used for processing digital data.
- **Voltage Regulator:** Maintains a stable voltage supply to the circuit components.
- **ADC0808**: Converts analog output of LM35 into digital format readable by the microcontroller.
- **7-Segment Displays:** Visual output devices used to display numerical data such as temperature readings.
- **Temperature Sensor:** Detects temperature changes and converts them into electrical signals.
- **Diodes:** Ensure current flows in one direction, protecting the circuit and enabling rectification.
- **Crystal:** Provides the clock frequency required for the microcontroller’s timing operations.
- **Push Buttons:** User inputs for resetting or controlling the system functions manually.
- **Relay:** Electrically operated switch used to control high-power devices with low-power signals.
- **Lamp:** Provides visual indication or lighting controlled by the system.
- **Buzzer:** Emits sound for warnings or notifications.

## 3.4 Algorithm

1. Read analog signal from MQ-2 gas sensor.
2. Convert analog signal to digital using ADC0808.
3. Process digital gas data using 89C51 microcontroller.
4. Compare gas level with threshold value.
5. Display gas status ("SAFE" or "LEAK DETECTED") on LCD.
6. Activate buzzer/LED if gas level > threshold.

## 3.5 Software Used

- Keil µVision (for programming 89C51)
- Proteus (for simulation)
- Arduino IDE (for flashing microcontroller)

# 4. IMPLEMENTATION

## 4.1 Hardware Assembly Steps

1. Microcontroller Setup (89C51): Fix the 8051 microcontroller onto a breadboard or microcontroller development board. Connect the crystal oscillator (11.0592 MHz or 12 MHz) to XTAL1 and XTAL2 pins with two 33pF capacitors to ground. Connect the crystal oscillator (11.0592 MHz or 12 MHz) to XTAL1 and XTAL2 pins with two 33pF capacitors to ground. Connect the 5V and GND from Arduino to Vcc and GND of the 8051 microcontroller. Interface push buttons to microcontroller input pins with pull-down resistors for user input (e.g., reset or control commands).
2. Connecting the MQ-2 Hydrogen Gas Sensor: Attach the gas sensor (e.g., MQ-2) to the analog input (via ADC0808 if used), and provide power and ground from Arduino.
3. Analog to Digital Conversion with ADC0808: The analog signal from LM35 is sent to the ADC0808, which requires a clock signal for proper functioning. This is generated using an RC oscillator or a 555 timer in astable mode. The ADC0808 converts the analog voltage to an 8-bit digital value, which is then sent to Port 1 of the 89C51 microcontroller.
4. Interfacing the 16x2 LCD Display: The Connect the 7-segment display or LCD to the I/O ports of the 8051 for temperature or status display (use resistors if needed).
5. Adding the Buzzer for Alerts: Wire LEDs and buzzer to output pins of 8051 through current-limiting resistors or transistors to indicate system status or alerts.
6. Power Supply Using Arduino Uno: Arduino's 5V and GND pins to power the 8051 microcontroller and other 5V components.

7. The 5V output powers the LCD, ADC0808, temperature sensor, lamp, relay and buzzer.

## 4.2 Software Code Overview

The program for the 89C51 is written in embedded C using the Keil µVision IDE. The key functions include:

- Initializing the LCD display.
- Triggering and reading values from ADC0808 connected to the gas sensor.
- Converting the ADC digital value to corresponding gas concentration levels.
- Displaying the gas concentration on the LCD.
- Activating the buzzer and/or alarm if the gas concentration exceeds the preset safe threshold.

## 4.3 Integration Process

Once all the individual modules (gas sensor, ADC, LCD, buzzer) are tested independently, they are integrated step-by-step:

1. Sensor and ADC Check: Confirm that the ADC accurately converts gas sensor output to digital format using LEDs or serial output for verification.

2. Microcontroller and LCD: Program the 89C51 to show a basic message on the LCD to ensure communication.

3. Full Integration: Combine the gas sensor-ADC module with the microcontroller and LCD. Display the real-time gas concentration levels.

4. Threshold and Buzzer Logic: Incorporate the condition to trigger the buzzer when gas concentration exceeds a preset safe threshold.

5. Final Testing: Power on the complete setup and verify the output across different gas concentration levels.

# 5. RESULTS AND DISCUSSION

## 5.1 Project in Working Condition (Screenshots/Images)

Below are images showcasing the successful implementation of the project:

**Image 1**: Full hardware setup showing the 89C51 microcontroller, MQ-2 sensor, ADC0808, 16x2 LCD, buzzer, and MP1584 power supply module.

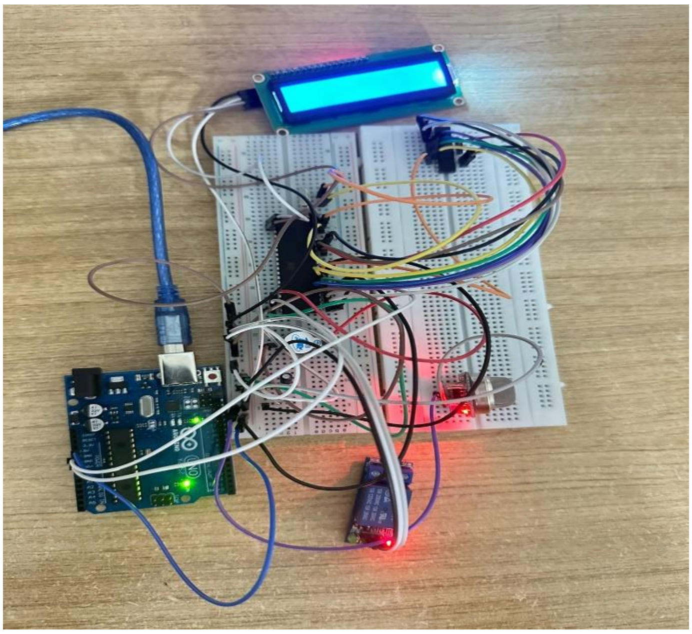


**Image 2:** LCD display showing real-time temperature

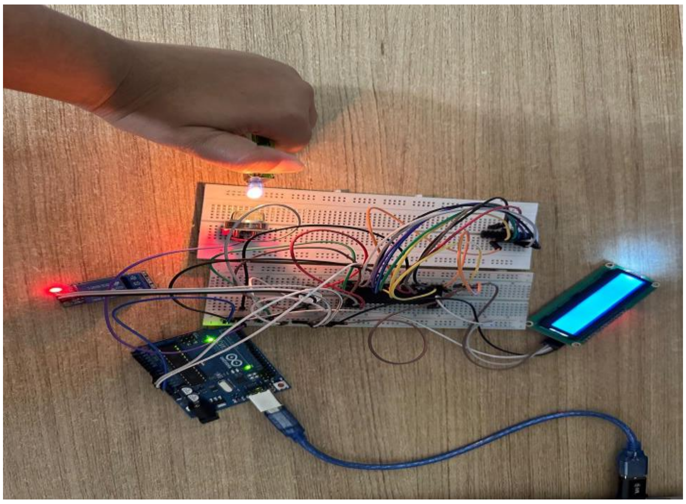


**Image 3:** Proteus image before detection of gas leakage.

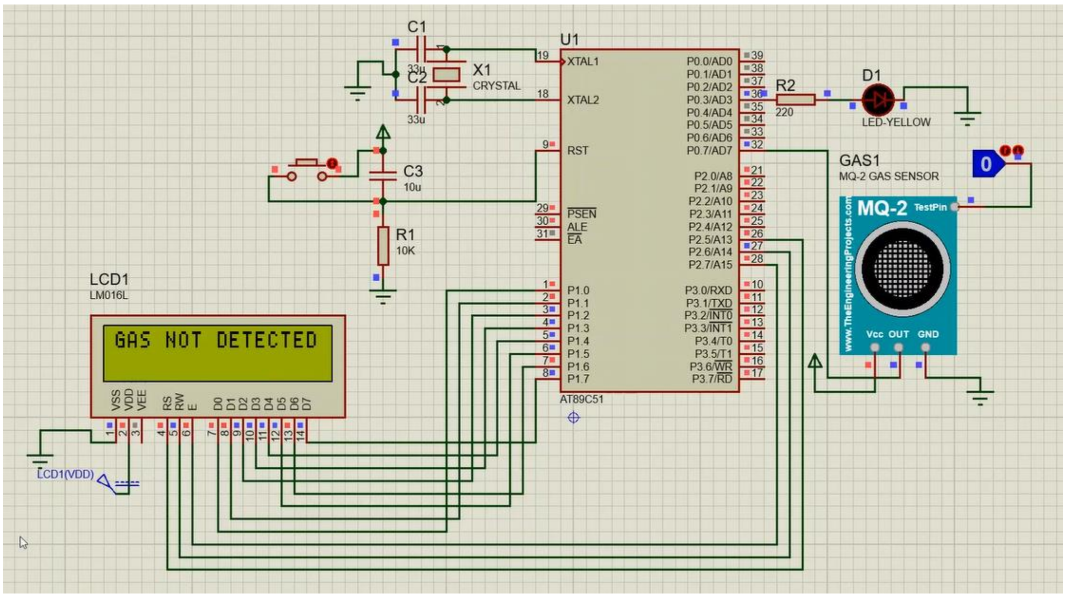
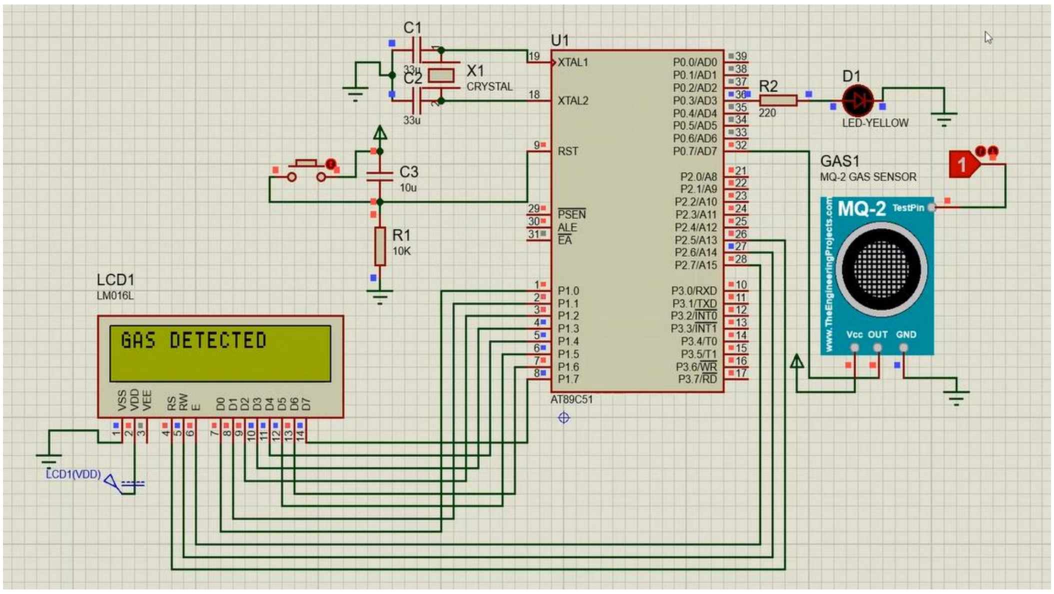



**Image 4:** Proteus image after detection of gas leakage.




## 5.2 Test scenarios

| Test No. | Condition | LCD output | Buzzer or led |
|---|---|---|---|
| 1 | Clean air | Not detected | OFF |
| 2 | Near LPG lighter | Detected | ON |
| 3 | Ventilated area | Not detected | OFF |

## 5.2Analysis

1. MQ2 sensor responds well to gas presence.
2. LCD provides real-time feedback.
3. Buzzer effectively warns on threshold breach.

# 6. EXPENDITURE DETAILS

| Component | Qty | Unit Cost (₹) | Total (₹) |
|---|---|---|---|
| 8051 Microcontroller | 1 | 80 | 80 |
| MQ2 Sensor | 1 | 120 | 120 |
| LCD Display (16x2) | 1 | 170 | 170 |
| Passive Buzzer | 1 | 10 | 10 |
| Breadboard & Wires | 1 set | 100 | 100 |
| Arduino uno | 1 | 450 | 450 |
| Total | | | 930 |

# 7. TIME SCHEDULE (GANTT CHART OR TABLE)

| Week | Activities Planned | Status |
|---|---|---|
| 1 | Problem identification & literature survey | Completed |
| 2-3 | Design and component selection | Completed |
| 4-5 | Hardware assembly | Completed |
| 6-7 | Coding and debugging | Completed |
| 8 | Testing and documentation | Completed |

# 8. Contributions

## a. Individual Contributions

| Member | Role/Contribution |
|---|---|
| 1. Ananya Deepak (1JT23EC014) | Components gathering, information |
| 2. B Swarthik (1JT23EC016) | Coding, Simulation |
| 3. Bhavani B (1JT23EC019) | Circuit design, Hardware connection |
| 4. Deeksha S (1JT23EC026) | Documentation, Presentation |

## b. Team contribution

**Joint Decision-Making**

From the very beginning of the project, all major decisions—such as the selection of components (89C51, LM35, ADC0808, LCD, MP1584), design of the circuit, and the programming approach—were made collectively. Each team member contributed ideas and insights, and decisions were finalized through consensus, ensuring everyone was aligned and understood the overall project direction.

**Regular Team Meetings**

To keep the project on track, the team held regular meetings throughout the development cycle. These meetings allowed us to review progress, assign tasks, share learnings, and resolve any obstacles. Whether in person or virtually, consistent communication helped maintain momentum and accountability.

**Collaborative Debugging and Testing**

Hardware troubleshooting and software debugging were approached as a team. For instance, when the LCD initially failed to display data, multiple members collaborated to trace wiring issues and adjust timing delays in the code. Similarly, during ADC integration and buzzer threshold testing, the group worked together to isolate and resolve logic and hardware errors. This collaborative effort strengthened our understanding and reduced overall development time

# CONCLUSION

The gas leakage detection system successfully detects LPG or methane and alerts the user via buzzer and LCD. It's cost-effective and suitable for home or lab use.

**Limitations:**

No data storage

Fixed threshold in code

Future Enhancements:

Add Wi-Fi or GSM alerts

Adjustable threshold via buttons

Use enclosure for safety

# REFERENCES


Mazidi, M.A. et al., The 8051 Microcontroller and Embedded Systems

Keil Microcontroller Development Tools (https://www.keil.com)

TutorialsPoint : Interfacing MQ2 with 8051 Microcontroller.